

condensation water leaking into the actuator.

Valve Flow Balancing:

1. Balancing & Control: Balancing should be accomplished by the Diaphragm and Control should be taken care by Actuator receiving signals from Room Thermostats or BMS.
2. Manual Override facility shall be provided to either open or close the valve.
3. Flow Setting Balancing (Commissioning) for the Valves should be simple and not require measuring devices.
4. Proper operation of the valve should not be dependent on additional operations like de-airing of the valve or flushing procedures
5. For higher pipe sizes 2-Way modulating valve with Equal percentage characteristic shall be used. This shall work on PID inputs and control the water flow rate as per the temperature requirements

Y-TYPE STRAINER (adhering to IS 12785)

The strainer element shall allow for cleaning without being dismantled. Flanges as per BS 10 shall be provided at inlet & outlet of connections. The connections shall be flanged only. The body shall be pressure tested at 1.5 times the operating pressure in the system. Strainers shall have SS 316 screen with 40 mesh perforations. Free screening area shall be min. 3 to 4 time's pipe cross section. All strainers shall be PN-16 rating and provided with equal size isolating valves so that the strainer may be cleaned without draining the system. Screen shall be removable and replaceable without disconnection of the main pipes.

Materials of Construction

(a)	Body	:	CI (ASTM 126) / MS (ASTM A536)
(b)	Filter/screen	;	SS 316
(c)	Rating (bar)	:	PN16
(d)	Nut-bolts/Fasteners	:	Galvanised Iron (GI)

INSULATION

Duct Thermal Insulation:(With Al Foil – Non Exposed)

- Insulation material should be Closed Cell Elastomeric Nitrile Rubber.
- The Aluminum foil (Embossed) shall be of 60 microns thickness and weight shall be 207 gsm as per ISO 2286-(P-2) 2016.
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 10°C as per (Tested acc. To ASTM C 518)
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 60,000 with tough Alu foil covering.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the

Building Standards (Scotland) Regulations 1990.

- The insulation material shall be FM (Factory Mutual) global, USA Approve Unlike other generic materials it shall be self-extinguishing, does not drip and does not spread flames.
- Base Material shall be free from CFC/HCFC.
- Density of material is 40-60 Kg/m³ in accordance to ASTM-D-1056.
- The insulation material should have service temperature range -20°C to +85°C.
- The insulation material shall be ROHS (Resistance of Hazardous Substance) complaint.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- The insulation material shall be installed as per manufacturer's recommendation.
- Adhesive shall be of low VOC to meet green building norms.
- **Material Color: Black**

Comply



Summary of Duct Insulation Thickness

	Insulation Thickness(mm)	Material to be used
Supply Air Duct	19	19 mm thick Closed cell elastomeric foam made from Nitrile rubber with tough Alu Foil.
Return Air Duct	13	13 mm thick Closed cell elastomeric foam made from Nitrile rubber with tough Alu Foil.

Comply

DUCT THERMAL INSULATION: (EXPOSED TO ATMOSPHERE)

- Insulation material technical specification – **Closed Cell elastomeric foam made up of nitrile rubber** with laminated layer of Glass Cloth.
- Insulation material should be of Closed Cell Elastomeric Nitrile Rubber.
- **Exposed side of insulation should have factory reinforced chemically treated glass cloth having thickness 7mil and weight 200 - 220GSM.**
- **Additional one layer of UV protection paint shall be applied for exposed ducts over GC Cloth**
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 10°C as per EN 12667.
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 11,000 with GC face barrier.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- Material shall have flammability class V-0 in accordance to UL94
- The insulation material shall be FM (Factory Mutual) global, USA Approve Unlike other generic materials it shall be self-extinguishing, does not drip and does not spread flames.
- Base Material shall be free from CFC/HCFC.
- Density of material is 40-60 Kg/m³ in accordance to ASTM-D-1056.
- The insulation material should have service temperature range -40°C to +105°C.

- The insulation material shall be ROHS (Resistance of Hazardous Substance) complaint.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Thickness of the insulation shall be as specified for the individual application.
- The insulation material shall be installed as per manufacturer's recommendation.
- Adhesive shall be of low VOC to meet green building norms.
- **Material Color: Black**

Comply

 Summary of Duct Insulation Thickness

	Insulation Thickness(mm)	Material to be used
Supply Air Duct	25	25 mm thick Closed cell elastomeric foam made from Nitrile rubber with a layer of factory laminated of GC.UV Paint shall be applied over GC
Return Air Duct	19	19 mm thick Closed cell elastomeric foam made from Nitrile rubber with a layer of factory laminated of GC. UV Paint shall be applied over GC



Comply

Duct Acoustic Insulation:

- Insulation material technical specification – **EPA (Environmental protection Authority) approved Open Cell elastomeric foam made up of nitrile rubber**
- Summary of Insulation thickness

Insulation Material Technical Specification – EPA approved Open Cell elastomeric foam made up of nitrile rubber

Insulation material shall be EPA Approved **Open Cell Elastomeric Nitrile Rubber with Anti-microbial**

Material shall be **EPA approved anti-microbial** and shall be **anti-bacterial** in accordance to ASTM E2180.

Material shall be Anti-fungal in accordance with ASTM G21. It shall also **pass fungal & bacterial resistance** test in accordance with **ISO 846**.

The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame,

Insulation material shall be bonded with the ducts using adhesive to the GI sheet. Insulation materials brought on site should be installed following the manufacturer recommended methodology and using the appropriately suitable manufacturer recommended cutting, measuring, pasting tools & tackles. It is advised to use the manufacturer's tool box for installation of insulation materials on site.

Thickness of the insulation shall be of 20mm Thick.

Density of material shall be 180-220 Kg/cum in accordance to ASTM-D-1667.

Material should have service temperature range -20°C to +85°C.

Comply



NRC (Noise Reduction Co-efficient) Value of material shall not be less than be 0.55 as per ISO: 354/ASTM 423C.

Freq(Hz)	100	125	160	200	250	315	400	500	630
Absorption Coefficient	0.01	0.03	0.04	0.05	0.09	0.16	0.20	0.32	0.45

Freq(Hz)	800	1000	1250	1600	2000	2500	3150	4000	5000
Absorption Coefficient	0.63	0.82	0.93	0.95	0.92	0.89	0.84	0.82	0.83

Under Deck Insulation:

Material: Closed cell Class O Nitrile Rubber.

Density: 40-60 Kg/M³

Insulation Thickness: 50 mm

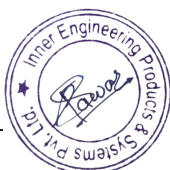
Facing: Aluminum Foil

- Insulation material should be Closed Cell Elastomeric Nitrile Rubber.
- The Aluminum foil (Embossed) shall be of 165 microns thickness and weight shall be 235 gsm as per ISO 2286-(P-2) 2016.
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 10°C as per EN 12667.
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 11,000 with external barrier facing .
- Material R-value shall be 8.1 hr ft² F/BTU at 10 °C.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- The insulation material shall have fire performance of V-0, HB as per UL 94, 1996.
- The insulation material shall be FM (Factory Mutual) global, USA Approve Unlike other generic materials it shall be self-extinguishing, does not drip and does not spread flames.
- Base Material shall be free from CFC/HCFC.
- Density of material is 40-60 Kg/m³ in accordance to ASTM-D-1056.
- The insulation material shall have service temperature range of -40°C to +105°C .
- The insulation material shall be ROHS (Resistance of Hazardous Substance) complaint.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Adhesive shall be of low VOC to meet green building norms.
- **Material Color: Black**

✚ Application Procedure:

- Clean the surface to be insulated.
- Providing and fixing 50 mm thick class 0 nitrile rubber in single layer and self adhesive Aluminum foil with the help of adhesive, G.I. cross wire, screw- washers & complete as per guidelines given in application manual.

Comply



- **Application Design:**



ACOUSTIC LINING WALL:

- Insulation material technical specification – **EPA approved Open Cell elastomeric foam made up of nitrile rubber** with separate layer of with GI frame grid and covered with 22G and 120 GSM for perforated GI sheet
- Summary of Insulation thickness

Insulation Material Technical Specification – EPA approved Open Cell elastomeric foam made up of nitrile rubber with GI frame grid and covered with 22G and 120 GSM for perforated GI sheet

Insulation material shall be EPA Approved **Closed Cell Elastomeric Nitrile Rubber with Anti-microbial**

Material shall be **EPA approved anti-microbial** and shall be **anti-bacterial** in accordance to ASTM E2180.

Material shall be Anti-fungal in accordance with ASTM G21. It shall also **pass fungal & bacterial resistance** test in accordance with **ISO 846**.

The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame

NRC (Noise Reduction Co-efficient) Value of material shall not be less than 0.55 as per ISO: 354/ASTM 423C.

Unlike other generic materials it should be self-extinguishing, does not drip and does not spread flames

Density of material shall be **180-220 Kg/cum x 20 mm** thickness in accordance to ASTM-D-1667.

Material should have service temperature range -20°C to 85°C.

Comply



Sound Absorption Coefficient of the material at 1/3 Octave Frequency in accordance with IS: ISO: 354/ASTM 423C is given below.

Freq(Hz)	100	125	160	200	250	315	400	500	630
Absorption Coefficient	0.01	0.03	0.04	0.05	0.09	0.16	0.20	0.32	0.45

Freq(Hz)	800	1000	1250	1600	2000	2500	3150	4000	5000
Absorption Coefficient	0.63	0.82	0.93	0.95	0.92	0.89	0.84	0.82	0.83

WALL ACOUSTIC INSULATION (ABOVE FALSE CEILING AT STORE NEAR MEETING ROOM)

- Following layers of acoustic insulation material which has **total thickness of 56 mm shall be fixed** with the help of necessary mechanical accessories.
- Layer 1: **Sound barrier is 3 mm thick** highly visco-elastic synthetic soundproofing membrane **having medium density 900-1100 Kg/m³** with high sound isolation property. It has minimum STC (Sound Transmission Class) value **20 dB** in accordance with ISO: 10140/ASTM E-90 (Based on Simulation).
- Transmission loss at different frequencies is in accordance **with ISO 10140(Part-III)/ASTM E-90.**

Frequency (Hz)	100	125	160	200	250	315	400	500
Air bone sound insulation Index (dB)	8.8	9.6	10.7	12	13.3	14.8	16.4	18
Frequency (Hz)	630	800	1000	1250	1600	2000	2500	3150
Air bone sound insulation Index (dB)	19.7	21.3	23.1	24.8	26.5	28.3	30.1	31.9

- It shall have minimum tensile strength 10 kgf/cm² in accordance with testing standard ASTM C 165-2012.
- It shall have minimum 30% Elongation at break in accordance with testing standard ASTM D412.
- It shall be pass flammability test in accordance with testing standard FMVSS 302 / IS 15061.
- Layer 2:
 - Material: Resin bonded mineral wool insulation
 - Density: 140 Kg/M³
 - Insulation Thickness: 50 mm
 - Facing: plain
- Thermal Conductivity of Resin Bonded slab shall be 0.037 W/m. K at an average mean temperature of 35 deg C in accordance with the testing standard ISO 8301

- It should have NRC (Noise Reduction Co-efficient) value of 0.90 at one third octave frequencies – with 25 mm Air gap as per IS: 8225 - 1987, “Measurement of Sound Absorption Coefficient in Reverberation Room” Equivalent to ISO: 354-2003 & ASTM 423-90

Frequency (Hz)	Sound Absorption Coefficient (α)
125	0.29
250	0.64
500	1.00
1000	1.00
2000	1.00
4000	1.00

- The insulation material shall have Class A Fire Performance Grade
- Insulation material shall confirm to NFPA 90A, 90B and LIFE SAFETY CODE NFPA101
- Insulation material shall confirm to ASTM E84 for surface burning characteristics of building materials
- This insulation material is to be encapsulated especially for an acoustical application
- Size: 1000 mm x 500 mm
- Confirming Standards: IS 8183
- The insulation material should be Incombustible (AS PER IS:8183/IS:3144) and moisture resistant & also it should not cause or accelerate any type of corrosion

Layer 3: Sound barrier is 3 mm thick highly visco-elastic synthetic soundproofing membrane **having medium density 900-1100 Kg/m³** with high sound isolation property. It has minimum STC (Sound Transmission Class) value **20 dB** in accordance with ISO: 10140/ASTM E-90 (Based on Simulation). Transmission loss at different frequencies is in accordance **with ISO 10140 (Part-III)/ASTM E-90.**

Frequency (Hz)	100	125	160	200	250	315	400	500
Air bone sound insulation Index (dB)	8.8	9.6	10.7	12	13.3	14.8	16.4	18
Frequency (Hz)	630	800	1000	1250	1600	2000	2500	3150
Air bone sound insulation Index (dB)	19.7	21.3	23.1	24.8	26.5	28.3	30.1	31.9

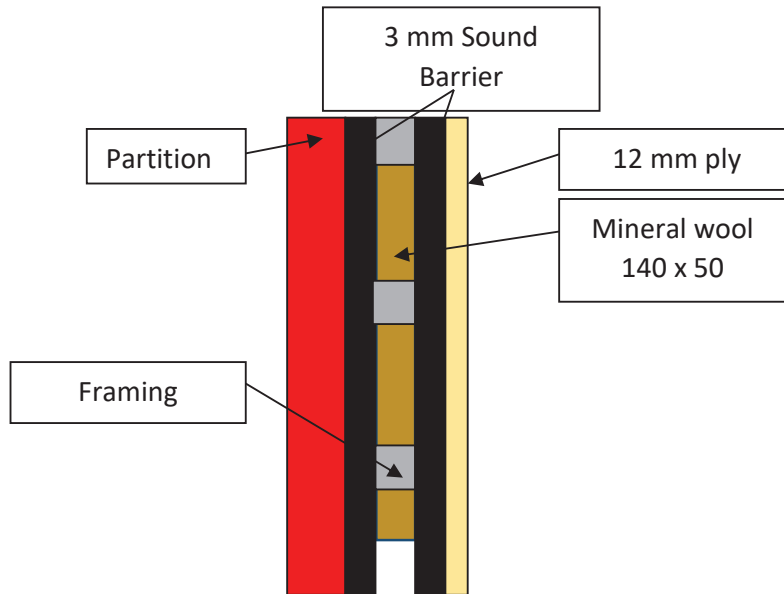
Application Procedure

- Clean the surface to be insulated.
- Providing & Fixing sound barrier having density of 900-1100 Kg/m³ and 3 mm thickness with high sound isolation property directly on brick wall above false ceiling area with the help of adhesive and mechanical accessories.
- Mineral wool having density 140 kg /m³ and 50 mm thickness shall be fixed with the help of framing which will fix with the help of mechanical accessories.
- Providing & Fixing sound barrier having density of 900-1100 Kg/m³ and 3 mm thickness with high sound isolation property with the help of adhesive and mechanical accessories.
- 12 mm ply shall be applied over framing as a finishing layer

Comply

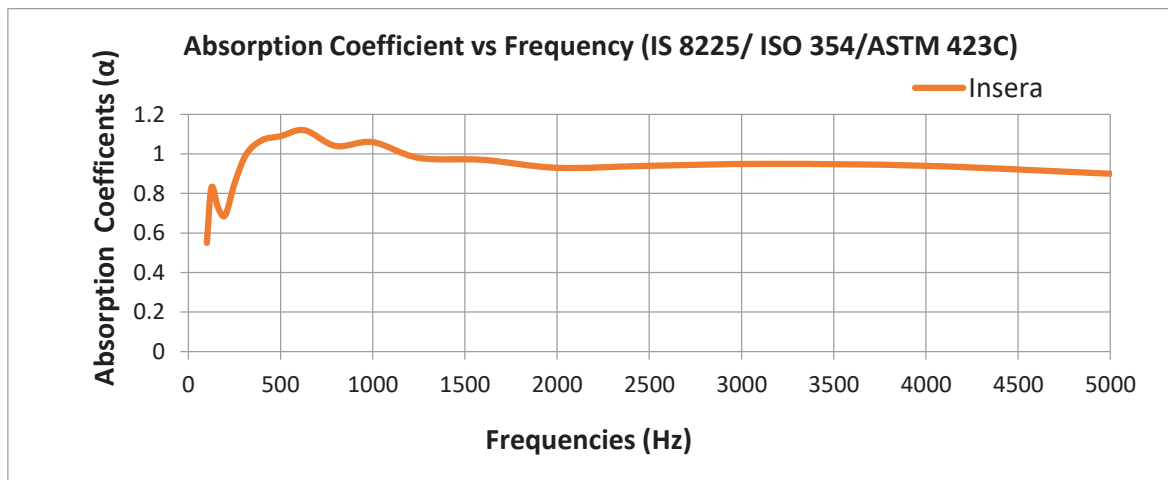


Application Design

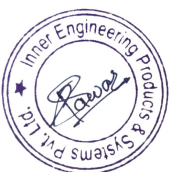


FALSE CEILING ACOUSTIC INSULATION (AT STORE NEAR MEETING ROOM)

- One layer of acoustic insulation material which has **total thickness of 50 mm** shall be placed over false ceiling.
- **Layer 1: Ceramic wool Encapsulated Insulation Pads** is Ceramic wool **50 mm** thick packed in Readymade machine stitched bad with one side loop and hook fastener has minimum **NRC (Noise Reduction Coefficient) value 0.90** in accordance with IS 8225/ ISO 354/ ASTM 423C. It has density **64 Kg/m³** and confirms to IS: 14656/IS:15402. It confirms to NFPA 90A, 90B, & Lift Safety Code NFPA 101. Sound absorption coefficients in octave band frequency are as below.



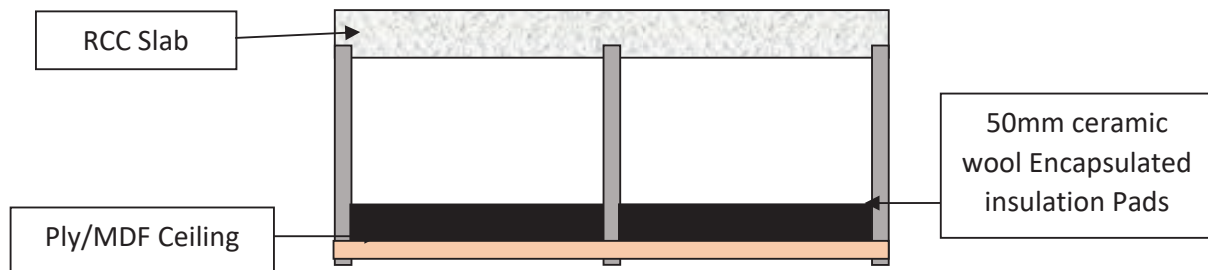
Comply



- **Application Procedure:**

- Clean the surface to be insulated.
- 50 mm thick ceramic wool encapsulated insulation pads shall be placed on false ceiling.

- **Application Design**



CHILLED WATER PIPING INSULATION:

- Insulation material technical specification – **EPA approved Closed Cell elastomeric foam made up of nitrile rubber** with separate layer of 26G AL Cladding.
- Insulation material should be Closed Cell Elastomeric Nitrile Rubber.
- The insulation shall be endosed with 26G Aluminium cladding, having tongue-in-groove joints held with SDST screws in a manner to shed water. The overlap of the sheet shall be sealed with a felt to prevent water ingress.
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 10°C (Tested acc. To ASTM C 518).
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 7,000 without any external barrier as per EN 12086.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- The insulation material shall have fire performance of V-0, HB as per UL 94, 1996.
- The insulation material shall be FM (Factory Mutual) global, USA Approve Unlike other generic materials it shall be self-extinguishing, does not drip and does not spread flames.
- Base Material shall be free from CFC/HCFC.
- Density of material is 40-60 Kg/m³ in accordance to ASTM-D-1056.
- The insulation material shall have service temperature range of -40°C to +105°C .
- The insulation material shall be ROHS (Resistance of Hazardous Substance) complaint.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Thickness of the insulation are mentioned below.
- The insulation material shall be installed as per manufacturer's recommendation.
- Adhesive shall be of low VOC to meet green building norms.
- **Material Color: Black**

Comply



Summary of Pipe Thermal Insulation Thickness Chart & Application:

Inner Pipe Dia. (NB)	Required Thickness	1st Layer thick(mm)	2nd Layer thick(mm)	Class 'O' Nitrile Rubber with separate layer of GC
25	32	32	-	Closed cell anti-fungal & anti-bacterial elastomeric foam made from nitrile rubber with 26G AL Sheet cladding
32	32	32	-	
40	32	32	-	
50	32	32	-	
65	38	38	-	Closed cell anti-fungal & anti-bacterial elastomeric foam made from nitrile rubber with separate layer of 26G AL Sheet cladding
80	38	38	-	
100	38	38	-	
125	44	44	-	Closed cell anti-fungal & anti-bacterial elastomeric foam made from nitrile rubber with separate layer of 26G AL Sheet cladding
150	44	44	-	
200	50	50	-	Closed cell anti-fungal & anti-bacterial elastomeric foam made from nitrile rubber with separate layer of 26G AL Sheet cladding
250	50	50	-	
300	50	50	-	
350 to 700	50	50	-	
750	64	32	32	Closed cell anti-fungal & anti-bacterial elastomeric foam made from nitrile rubber with separate layer of 26G AL Sheet cladding

Noted

PASSIVE FIRE PROTECTION SYSTEM

(A). Intumescent coating Fire stopping System:

This system is to be used for the sealing of service shaft like plumbing, firefighting and electrical shafts openings in the Wall penetration.

Technical Specification of Composite Board for Wall Openings

Fire barrier for sealing of wall openings application with a Intumescent coating Fire stopping System used in combination with Mineral Wool Board tested in accordance to EN 1366-3, ETA-16/0523 standards and capable of providing a fire resistance up to 2 hrs. The product along with the system should be designed for use with single cables, cable bundles, cable trays, combustible and non-combustible pipes, fire dampers and also insulated ventilation systems to prevent the spread of smoke, fire and heat.

Application Details in accordance EN 13501 for water-based hybrid firestop coating on mineral wool board of minimum density 140 Kg/M3, melting point > 1000 Deg C.